

Efficient and High Fidelity Multimodal Machine Unlearning for Information With Deep Memory

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Abstract—Multimodal large models perform exceptionally good at complex tasks due to learning from vast amounts of data. To comply with the “right to be forgotten”, and teach multimodal models to forget sensitive information, an efficient multimodal large model unlearning is required. Compared to existing works, in this article, we focus on improving unlearning efficiency and preserving the model fidelity. Instead of generating meaningless outputs, a multimodal model can generate meaningful outputs without sensitive information that is requested to be forgotten. Our main idea is to exploit and only update model parameters that contribute to learning sensitive information. In addition, to improve the model’s unlearning ability, we make a multimodal model to unlearn information that is with a deep memory. Specifically, we enhance the unlearning strategy by designing and using “negative” samples (which are paired samples to samples asked to be forgotten) to confuse the model. Evaluations on vision-language tasks, such as Visual Question Answering and Image Captioning, using multiple datasets, demonstrate the efficiency and fidelity of our unlearning strategy. Compared to existing methods, our method obtains an average of $68\times$ improvement. Meanwhile, the average utility-forget balance score has improved $6\times$ on average and reached up to $70\times$, across multiple models and datasets.

Index Terms—Machine unlearning, privacy protection.

I. INTRODUCTION

MULTIMODAL large language models (MLLMs) have demonstrated remarkable capabilities across tasks such as image-text understanding, cross-modal generation, and human-computer interaction. By leveraging vast amounts of multimodal data, the models learn complex correspondences between heterogeneous modalities such as vision and language, enabling remarkable proficiency in diverse applications. However, the widespread deployment of MLLMs has raised growing concerns about data privacy and security [1], [2], [3]. The training data may contain sensitive user information such as facial images, real names, and contact details. If MLLMs learn too much from the training datasets, it appears to pose a threat to privacy leakage and regulatory violations. In response, machine unlearning [4] has emerged as a new learning paradigm to

comply with privacy regulations (GDPR [5]), which grant individuals “the right to be forgotten”. Machine unlearning techniques enable users to request the deletion of their personal data or specific information from models. That is, machine unlearning techniques make models perform close to or exactly become the ones that are retrained without the data samples that are requested to be forgotten.

In the context of classification model unlearning, some approaches achieve unlearning by sharding datasets and only training sub-models based on the shards [4], [6]. The sub-models are then aggregated to form the entire model. Once users initiate a request to forget some data samples, the corelative sub-models whose training data involve the requested samples are retrained, thereby improving efficiency. When it comes to transformer-based models [7], especially generative AI models which exhibit remarkable generation capabilities, MLLMs, large language models, the unlearning demand is high and urgent. This is because large models learn a large amount of data and may unintentionally output sensitive personal information such as names and email addresses. In LLMs, existing works achieve unlearning by optimizing all model parameters [8], [9]. More concerningly, MLLM unlearning can lead to significant costs due to retraining on massive multimodal datasets and their complex model architectures. Recent multimodal model unlearning methods improve efficiency by decoupling the dependencies of modalities (text and image) for data samples requested to be deleted [10]. Afterwards, the model is trained on a small subset of the remaining dataset to preserve model accuracy. However, existing methods have several limitations in practical applications. Most unlearning methods focus on data-level rather than information-level. Some require updating all model parameters to remove sensitive information from regular samples, resulting in high computational overhead, while some have incoherent and even meaningless outputs, which compromise output reliability and user experience.

In this paper, we propose an Efficient Multimodal model Machine Unlearning, called EMMU. The EMMU framework efficiently achieves sensitive information-level unlearning. Specifically, when multimodal data samples (such as images and texts) contain sensitive information such as user names and addresses, our goal is to make multimodal models efficiently forget the sensitive information. In addition, multimodal models may memorize some samples deeper than others, especially those involving celebrities, making such sensitive information harder to forget and increasing privacy risks. We define data samples with deep memory as *hard samples*.

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Considering the key challenges of MLLMs in privacy-preserving unlearning, our research focuses on two core problems: 1) How to achieve efficient basic unlearning strategies for regular sensitive samples. 2) How to address hard samples that the model memorizes more strongly.

Our main idea is to locate crucial parameters strongly correlated with sensitive information, combined with lightweight parameter update mechanisms to achieve sensitive information unlearning. We introduce a memory degree evaluation using negative samples and an enhanced unlearning strategy with adaptive parameter selection and updates. The negative samples are paired with the original samples, but have different meaning. For example, we can generate negative samples by using images of others but labeled as Donald Trump, or using images of Donald Trump but named them as John M. Welsh.

The main contributions of this paper are summarized as follows:

- 1) We propose a basic unlearning strategy for multimodal models that efficiently removes sensitive information from regular samples by locating and updating parameters highly correlated with target sensitive information, significantly improving unlearning efficiency while preserving model performance.
- 2) We develop a memory degree assessment mechanism to quantify the memory degree of different samples in a given model, and design an enhanced unlearning strategy for hard samples that exhibit high memory degree, improving model unlearning capability and overall robustness.
- 3) We propose EMMU, a comprehensive machine unlearning framework for multimodal models that combines both basic and hard sample unlearning strategies, and conduct extensive experimental validation on different multimodal models for tasks such as Visual Question Answering (VQA) [11] and Image Captioning [12]. The results demonstrate that EMMU outperforms existing multimodal unlearning methods in both effectiveness and computational efficiency.

II. RELATED WORK

A. Foundations of Machine Unlearning

Existing unlearning methods are generally classified into exact and approximate unlearning. Exact unlearning requires the unlearned model to be indistinguishable from one retrained from scratch. To reduce cost, SISA [4] partitions data into shards and slices to train multiple submodels, so only those containing forgotten samples are retrained. ARCANE [6] reduces retraining costs by ordering and sorting the data to be forgotten. However, as model and dataset scales grow, exact methods face prohibitive retraining overheads. Approximate unlearning relaxes this requirement, allowing bounded discrepancies between the unlearned and retrained models. These methods typically update model parameters directly. Some works provide certified guarantees [13], while others studies [14], [15] modify the model parameters by computing the Hessian matrix, which are inefficient for large models. Most existing approximate unlearning methods focus on supervised learning scenarios, with only a few studies

extending them to the field of reinforcement learning. TRA-JDELETER [16], as the first trajectory-level unlearning method for offline reinforcement learning agents, provides a solution to the privacy protection problem in offline reinforcement learning scenarios.

B. Unlearning in Large Language Models

In large language models, several studies focus on removing harmful knowledge to ensure safer output. For example, KUL [8] uses gradient ascent to maximize the loss function, causing the model predictions on specific samples to deviate from the original direction. LLMU [9] combines multiple loss functions, achieving unlearning while preserving the performance of the model on normal inputs. It is noteworthy that the FFN layers in transformers are recognized for capturing abstract knowledge [17], [18]. Based on this observation, researchers have proposed modifying privacy-related neurons in FFN layers to achieve machine unlearning [19], though this method does not truly “unlearn” data as it does not alter the model parameters.

C. Unlearning in Multimodal Large Language Models

Multimodal unlearning presents unique challenges due to the need to handle information across multiple modalities. MultiDelete [10] first addresses this problem by decoupling the association between image and text in the data to be forgotten, preserving single-modal representations while maintaining multimodal understanding of other data. EFUF [20] utilizes the CLIP [21] model to assess the consistency between text and images, adjusting the loss function to reduce hallucinations in the model’s output. However, these methods still require modifying all model parameters to achieve unlearning. This paper explores a more efficient cross-modal unlearning method.

III. PRELIMINARIES

A. Multimodal Machine Unlearning

Multimodal machine unlearning aims to selectively remove the influence of specific information from models that process multiple data modalities (image and text). This study focuses on vision-language models and specializes in VQA and Image Captioning tasks. Consider a vision-language model F_θ with parameters θ , trained on a dataset $\mathcal{D} = \{(I_k, T_k)\}_{k=1}^N$ containing N image-text pairs, where I_k and T_k represent the image and text for the k -th data pair, respectively. In the VQA task, $T_k = \{(Q_t, A_t)\}_t^{|T_k|}$ denotes the question-answer pairs associated with the image I_k , while in the Image Captioning task, T_k is the caption for the image I_k . The training objective is to minimize the following loss function:

$$\mathcal{L}(\theta) = \frac{1}{|N|} \sum_{k=1}^N \ell(F_\theta(I_k, T_k), T_k),$$

where $\ell(\cdot, \cdot)$ is the cross-entropy loss. Now, suppose we need the model to unlearn the influence of a subset of data $\mathcal{D}_u$ from the model F_θ . This subset is defined as $\mathcal{D}_u = \{(I_i^u, T_i^u)\}_{i=1}^{|\mathcal{D}_u|}$. Accordingly, we define $\mathcal{D}_r = \mathcal{D} \setminus \mathcal{D}_u$ as the remaining dataset

after removing $\mathcal{D}_u$. Ultimately, we aim to obtain a new model F'_θ that has completely eliminated the influence of $\mathcal{D}_u$ on the model. This means that the performance of F'_θ on $\mathcal{D}_u$ should be as if $\mathcal{D}_u$ had never participated in the training of F_θ . At the same time, we need to ensure that the performance of F'_θ on the test set $\mathcal{D}_t$ is as close as possible to that of F_θ .

B. Shapley Additive Explanations

To achieve the unlearning of sensitive information in the data by modifying only a portion of the parameters, we need to identify the crucial parameters in the model that are highly correlated with the data to be forgotten. In this paper, we adopt the SHapley Additive exPlanations (SHAP) [22] to evaluate the influence of parameters on model outputs. SHAP is a model interpretation method based on game theory, used to quantify the importance of each feature on model predictions. In the SHAP, the importance of each feature is measured by its Shapley value [23], which represents the average marginal contribution of the feature to the model's prediction. For a given model f and instance x , the Shapley value of feature i is defined as:

$$\phi(f, i) = \sum_{S \subseteq H \setminus \{i\}} \frac{|S|!(|H| - |S| - 1)!}{|H|!} (v(S \cup \{i\}) - v(S)),$$

where H is the set of all features, S is a subset not containing feature i , and $v(S)$ is the prediction value for the subset S . In our method, SHAP analyzes the contribution of intermediate layer outputs to final predictions, allowing us to identify the parameters that are most relevant to the data to be forgotten.

IV. SYSTEM MODEL AND THREAT MODEL

A. System Model

We consider a typical service scenario of MLLMs, where the system involves three main roles: the service provider, data owners, and users. The service provider is responsible for model training, deployment, and maintenance, and possesses both the model parameters and the training data. The model is typically deployed on servers and provides services through an interface. Users access the model via this interface, and some users may act maliciously to extract sensitive information from the model. Data owners are entities whose data are used in training and may issue requests to remove their data.

In this system, a request to forget can be initiated either by data owners or triggered by the service provider based on privacy risk detection. Once sensitive information requiring removal is identified, the service provider performs unlearning operations on the model to eliminate its memorization of the corresponding sensitive information.

B. Threat Scenario

MLLMs are typically trained on large-scale, multi-source datasets, which may contain sensitive information such as facial images, personal identities, and health records. Once such data is used for training, the model may retain implicit memorization of it, posing a risk of privacy leakage. Under the system setting described above, the adversary can interact with the model as

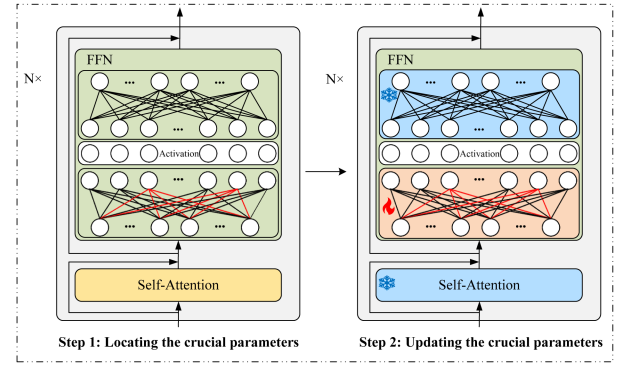


Fig. 1. EMMU comprises two key steps: crucial parameters localization and updating. We only update parameters highly associated with sensitive information in the unlearning data to achieve unlearning.

a malicious user and attempt to extract sensitive information stored in the model through carefully crafted prompts. To address this threat, multimodal machine unlearning aims to eliminate the model's memorization of sensitive information from its parameters without retraining the entire model, steering the model's behavior toward that of a model never trained on the target samples. Multimodal machine unlearning can mitigate privacy threats targeting training data, such as model extraction attacks and membership inference attacks (MIA), but may remain vulnerable to more sophisticated attack strategies.

C. Adversary's Goals and Capabilities

Adversary's goals. The primary objective of the adversary is to extract sensitive private information related to the training data from the model. To this end, the adversary can design targeted prompts and analyze the model's generated responses over multiple interaction rounds to infer sensitive information in the training data. For instance, by combining specific images with carefully crafted questions, the adversary may elicit the model into revealing identity or health-related information about particular individuals.

Adversary's capabilities. We adopt a black-box setting, in which the adversary can only interact with the model through a query interface, without access to internal parameters or gradients. Specifically, the adversary can submit image-question pairs and observe the model's outputs, but cannot obtain any information about the model architecture or training details. This assumption aligns with the typical deployment practice of current multimodal large language models.

V. METHODS

A. Method Overview

We propose EMMU, a unified machine unlearning framework for multimodal models, consisting of two complementary strategies: a basic strategy for regular sensitive samples and an enhanced strategy for hard samples. As shown in Fig. 1, EMMU involves two key steps: i) locating crucial parameters and ii) updating the crucial parameters. Unlike existing approaches,

EMMU only modifies a small subset of parameters, ensuring efficiency and fidelity.

For the basic strategy, we localize parameters by erasing the sensitive content in the input and comparing intermediate activations before and after erasure. Additionally, we use SHAP to attribute sensitive outputs to internal representations. We fuse these two perspectives to obtain a subset of crucial parameters. We then optimize only these parameters with a multi-objective loss that balances forgetting efficacy and model fidelity.

In the enhanced strategy, we cluster sensitive data to select representative samples for localization. We identify a set of crucial parameters using the basic strategy and further fuse them with parameters ranked by difference-based scores from lightweight fine-tuning with negative samples. During updates, a batch-aware mechanism modifies only the intersection of the fused crucial parameter set with the parameters activated by the current batch, improving stability and specificity.

This unified framework enables precise, efficient, and robust unlearning of sensitive information across varying data complexities.

B. Basic Strategy for Unlearning Sensitive Information

1) *Basic Strategy Description:* The basic strategy enables fine-grained unlearning of sensitive information from regular samples by identifying and updating crucial parameters highly correlated with the target sensitive information. Our objective faces two main challenges: i) how to locate crucial parameters related to sensitive information and ii) how to ensure the model fidelity while modifying parameters to achieve unlearning.

To address these challenges, we locate crucial parameters during inference using a collaborative decision-making mechanism and update them with a multi-objective optimization strategy. The results obtained during inference reflect the model's learned information and knowledge, making it an ideal stage for parameter localization. Based on previous works [17], [18], FFN layers in Transformer models can capture semantic information. Therefore, we perform the localization and updating of crucial parameters within FFN layers. However, we do not assume that sensitive information is exclusively stored in FFN layers. We focus only on the parameters of the last few layers of the model, as these layers are typically closely related to sensitive content. We locate crucial parameters mainly from two perspectives: i) parameters influenced by the input, identified by comparing intermediate activations before and after erasing sensitive information; and ii) parameters influencing the model outputs, identified using SHAP to attribute model outputs to internal representations. Parameters that are significantly affected by sensitive information or have the high-contribution scores are considered crucial. We fuse the parameters identified from both perspectives and update this crucial subset using a multi-objective optimization strategy that balances forgetting sensitive information with preserving overall model fidelity.

2) *Locating the Crucial Parameters:* We locate crucial parameters in the last few layers of the modality fusion module, which integrates information from different modalities and plays a key role in propagating sensitive information. To identify

parameters influenced by input, we compare intermediate activations when generating the first token for original and erased inputs, where the first token captures a condensed representation of global information. We also analyze representations when generating all tokens containing sensitive information and compare them with first-token representations from images without sensitive information for attribution.

Let the unlearning set size be B , with images $I^u = \{I_1^u, I_2^u, \dots, I_B^u\}$. We erase sensitive content to obtain $I^d = \{I_1^d, I_2^d, \dots, I_B^d\}$. We select half of the set, $\mathcal{D}_{u1}$ (where $|\mathcal{D}_{u1}| = B/2$), to locate crucial parameters. Fig. 2 illustrates the process of crucial parameters localization.

a) Parameters identification based on input sensitivity

We input both the original images containing sensitive information and the images with sensitive information erased into the model for inference to investigate the differences in model outputs under different inputs. This allows us to identify the parameters learned by the model that are related to sensitive information in the input images.

For the k -th sample, we extract the corresponding intermediate layer representations when generating the first token from data containing sensitive information. Similarly, we extract representations from data without sensitive information. In the l -th layer of the last few layers T_l , the output representation of the i -th neuron in the intermediate layer for the k -th sample containing sensitive information is denoted as $Ou_i^{l,k}$. Correspondingly, the output representation after erasing the sensitive information is denoted as $Od_i^{l,k}$.

To accurately identify the crucial parameter subset, we design an input sensitivity-based scoring mechanism to effectively capture the responsiveness of parameters to sensitive information. Specifically, for each neuron i in the l -th layer of the modality fusion module, we calculate its sensitivity score $score(l, i)$:

$$score(l, i) = \frac{1}{B/2} \sum_{k=1}^{B/2} |Ou_i^{l,k} - Od_i^{l,k}|.$$

This score reflects the average difference in the outputs of neurons under inputs with and without sensitive information, thereby quantifying each neuron's responsiveness to sensitive information. Based on the sensitivity of neurons to sensitive information, we further evaluate the relevance of their corresponding parameters to sensitive information in the input data. We select the parameters corresponding to the top R neurons ranked by sensitivity scores as candidate crucial parameter subset:

$$\mathcal{P}_{in}^l = \{W_i^l | TopK(score(l, i), R), l \in T_l\},$$

where W^l represents the parameter weights of the intermediate layer in the l -th layer. The layers closer to the output layer are more relevant to the task, so we select a fixed number of parameters related to sensitive information to ensure that crucial information is captured.

Finally, we combine the results to obtain the crucial parameter subset of the modality fusion module based on input sensitivity:

$$\mathcal{P}_{in} = \bigcup_{l \in T_l} \mathcal{P}_{in}^l.$$

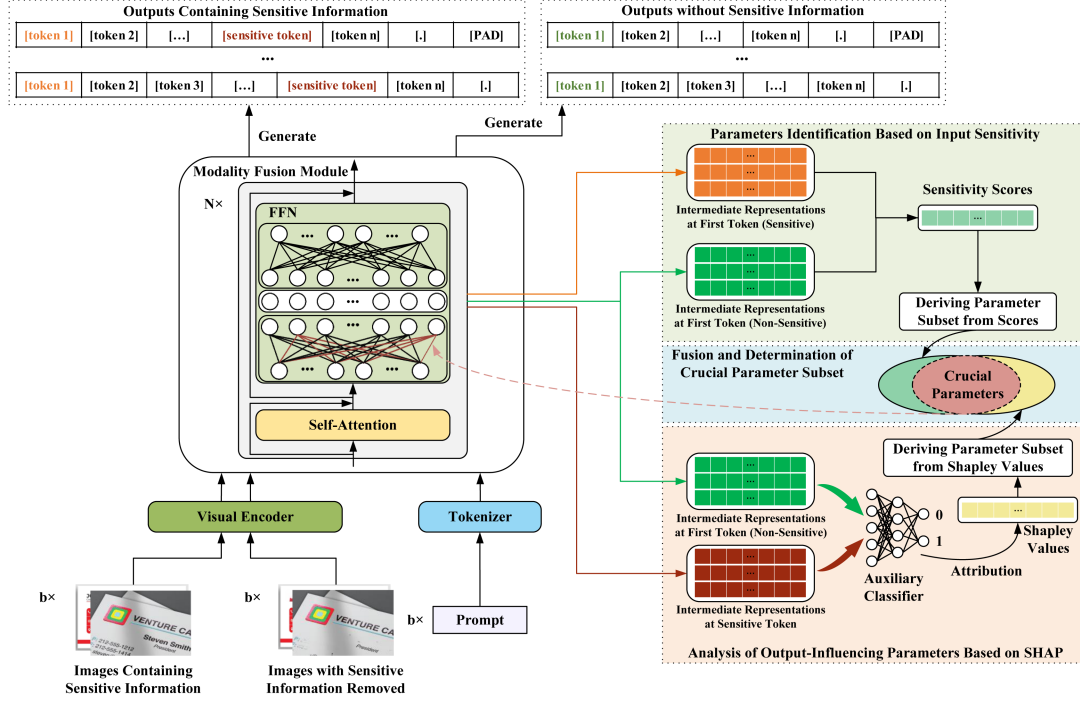


Fig. 2. The process of crucial parameters localization in basic unlearning strategy. This primarily includes: 1) parameters identification based on input sensitivity; 2) analysis of output-influencing parameters based on SHAP; 3) fusion and determination of the crucial parameter subset. In the VQA task, the “Prompt” corresponds to the question associated with the image data, while in the Image Captioning task, the “Prompt” refers to a cue phrase such as “A picture of.”.

b) Analysis of Output-Influencing Parameters Based on SHAP

For the analysis of parameters that influence the output of the model, our goal is to identify potential causes of the generation of tokens that contain sensitive information. We utilize the SHAP to analyze crucial parameters that affect the output of the model. The specific steps are as follows:

i) *Collecting intermediate layer representations*: We extract intermediate layer representations of the model during inference on images containing sensitive information and those without sensitive information to construct a dataset for training auxiliary classifiers. First, sensitive keywords are extracted from text descriptions of the data to be forgotten and converted into a token sequence K . Suppose the current inference batch size is b , and the maximum generated sequence length is N . We define a keyword mask matrix $M_{kw} \in \{0, 1\}^{b \times N}$, with its elements defined as follows:

$$M_{kw}[k, n] = \begin{cases} 1, & \text{if the } n\text{-th token of the } k\text{-th} \\ & \text{sample is in the keyword set;} \\ 0, & \text{otherwise.} \end{cases}$$

Let the number of sensitive tokens matched in the current batch be N_b . Using the keyword mask M_{kw} , we extract the intermediate layer representations in the l -th layer of the last few layers T_l when generating sensitive information, denoted as $O_s^{l,b} \in \mathbb{R}^{N_b \times d}$, where d represents the feature dimension of the intermediate layer output. By further aggregating all intermediate layer representations from the data to be forgotten, we obtain $O_s^l \in \mathbb{R}^{N_s \times d}$, where N_s is the total number of sensitive tokens matched in the data to be forgotten. Similarly, we extract

the intermediate layer representations for the first token generated during inference on images without sensitive information, denoted as $O_d^l \in \mathbb{R}^{B/2 \times d}$, where $B/2$ is the total number of non-sensitive samples. We assign a label of $\mathbf{1}$ to the intermediate layer representations containing sensitive information, O_s^l , and a label of $\mathbf{0}$ to those without sensitive information, O_d^l .

ii) *Training auxiliary classifiers*: We define the auxiliary model $f^l(\cdot)$ at the l -th layer as a simple neural network that takes the feature representation $x \in \mathbb{R}^{1 \times d}$ from the intermediate layer's output as input and outputs a prediction $y \in \{0, 1\}$, where $\mathbf{1}$ indicates containing sensitive information and $\mathbf{0}$ indicates not containing sensitive information. We use the intermediate layer representations O_s^l and O_d^l collected in i) as training data. At each layer l of T_l , we train a single auxiliary classifier specifically for that layer to determine whether the intermediate layer output contains sensitive information. The model optimization is carried out using the Adam [24] optimizer, with a cross-entropy loss function, enabling the auxiliary classifier to effectively learn features related to sensitive information from the intermediate layer representations.

iii) *Shapley values calculation*: We apply the SHAP method to analyze the contribution of input features to the auxiliary classifier's prediction of $\mathbf{1}$, thus calculating the Shapley value for the output of each neuron i . Considering that the data in the current scenario follows the same distribution, calculating the Shapley values for all generated labels containing sensitive information could be time-consuming. Therefore, we randomly sample m samples from $O_s^l \in \mathbb{R}^{N_s \times d}$, denoted as $O_{s'}^l \in \mathbb{R}^{m \times d}$, to reduce the computational cost. Using the SHAP explainer, we compute the Shapley value $\phi_i^{l,k}$ for the sampled k -th intermediate layer

representation:

$$\phi_i^{l,k} = \text{Explainer} \left(f^l(\cdot), O_s^{l,k} \right).$$

We aggregate the Shapley values across all samples to obtain the total contribution ϕ_i^l of each neuron's output to the classifier's prediction:

$$\phi_i^l = \frac{1}{m} \sum_{k=1}^m \phi_i^{l,k}.$$

iv) Parameters selection: Based on the contribution ϕ_i^l of each neuron's output to the classifier's prediction, we determine the relevance of the corresponding parameters to sensitive information. We select parameters corresponding to the top R neurons ranked by contribution as the candidate crucial parameter subset:

$$\mathcal{P}_{shap}^l = \{W_i^l | \text{TopK}(\phi_i^l, R), l \in T_l\}.$$

We merge the results to obtain the crucial parameter subset influencing model output:

$$\mathcal{P}_{shap} = \bigcup_{l \in T_l} \mathcal{P}_{shap}^l.$$

c) Fusion and determination of crucial parameter subset

To comprehensively consider the impact of sensitive information on the model, we fuse the two types of parameters based on input sensitivity and SHAP. Specifically, we take the intersection of crucial parameters identified in each layer of the model's fusion module, simultaneously considering the importance of these two types of parameters to the sensitive information to be forgotten. Formally, in the l -th layer, the crucial parameters are defined as:

$$\mathcal{P}_{cru}^l = \mathcal{P}_{in}^l \cap \mathcal{P}_{shap}^l.$$

Finally, we obtain the crucial parameter subset that needs to be modified for unlearning data in the current scenario:

$$\mathcal{P}_{cru} = \bigcup_{l \in T_l} \mathcal{P}_{cru}^l.$$

3) Updating the Crucial Parameters: After determining the crucial parameter subset that needs modification, we continuously update the previously identified crucial parameter subset $\mathcal{P}_{crucial}$ using the remaining half of the data $\mathcal{D}_{u2}$ from the unlearning set, where $\mathcal{D}_{u2} = \mathcal{D}_u \setminus \mathcal{D}_{u1}$. We then design a multi-objective optimization strategy to update these updated parameters. This strategy aims to balance three crucial objectives: unlearning sensitive information, maintaining overall performance, and preventing excessive unlearning. The process of updating the crucial parameters is shown in Algorithm 1. We will elaborate on the design of the optimization objectives in the following.

a) Unlearning sensitive information: To minimize the impact of sensitive information in the data to be forgotten on the model, we adopt a gradient ascent approach for optimization. For samples in the unlearning set $\mathcal{D}_{u2}$, we achieve unlearning

Algorithm 1: Updating the Crucial Parameters.

Input: original model F_θ , unlearned model F'_θ , unlearning set $\mathcal{D}_{u2}$, remaining set $\mathcal{D}_r$, target layers T_l , crucial parameter pool $\mathcal{P}_{cru} = \bigcup_{l \in T_l} \mathcal{P}_{cru}^l$, learning rate η
Output: unlearned model F'_θ

- 1: Initialize $F'_\theta = F_\theta$
- 2: Initialize the epoch's crucial parameters in the crucial parameter pool: $\mathcal{P}_E = \emptyset$
- 3: **for** each epoch $e \leftarrow 1$ **to** E **do**
- 4: **for** each batch $b \sim (\mathcal{D}_{u2}, \mathcal{D}_r)$ **do**
- 5: Initialize the batch's crucial parameters $\mathcal{P}_b = \emptyset$;
- 6: **for** each $l \in T_l$ **do**
- 7: Calculate the sensitivity score $score(l, i)$ for each neuron i in the intermediate layer using F'_θ ;
- 8: Select parameters $\mathcal{P}_{in}^l \leftarrow \text{TopK}(score(l, i), R)$ corresponding to the top R neurons;
- 9: **end for**
- 10: Update $\mathcal{P}_b \leftarrow \bigcup_{l \in T_l} \mathcal{P}_{in}^l$;
- 11: Calculate the parameters in the crucial parameter pool: $\mathcal{P}_{pool}^b \leftarrow \bigcup_{l \in T_l} (\mathcal{P}_{in}^l \cap \mathcal{P}_{cru}^l)$;
- 12: **if** $e == 1$ **then**
- 13: Set $\mathcal{P}_E^b \leftarrow \mathcal{P}_{pool}^b$;
- 14: **else**
- 15: Calculate the parameters that need to be updated in the crucial parameter pool:
 $\mathcal{P}_{upd}^b \leftarrow \mathcal{P}_E^b \setminus \mathcal{P}_{pool}^b$;
- 16: Update the crucial parameter pool:
 $\mathcal{P}_{cru} \leftarrow \mathcal{P}_{cru} \setminus \mathcal{P}_{upd}^b$;
- 17: Set $\mathcal{P}_E^b \leftarrow \mathcal{P}_{pool}^b$;
- 18: **end if**
- 19: Calculate the multi-objective loss:
 $\mathcal{L} = \gamma_1 \cdot \mathcal{L}_1 + \gamma_2 \cdot \mathcal{L}_2 + \gamma_3 \cdot \mathcal{L}_3$;
- 20: Update the crucial parameters of the unlearned model: $\theta_{\mathcal{P}_{cru}} \leftarrow \theta_{\mathcal{P}_{cru}} - \eta \nabla_{\mathcal{P}_{cru}} \mathcal{L}$;
- 21: **end for**
- 22: **end for**
- 23: **return** F'_θ

by maximizing task-specific loss:

$$\mathcal{L}_1 = -\frac{1}{|\mathcal{D}_{u2}|} \sum_{k=1}^{|\mathcal{D}_{u2}|} \ell(F'_\theta(I_k^u, T_k^u), T_k^u),$$

where $\ell(\cdot, \cdot)$ represents the cross-entropy loss function, (I_k^u, T_k^u) represents the image-text pairs in the unlearning scenario.

b) Maintaining overall model performance: To maintain the model's performance on non-sensitive data while achieving data unlearning, we minimize the distance between the model's distributions on the remaining set $\mathcal{D}_r$ before and after the unlearning:

$$\mathcal{L}_2 = \sum_{(I_k^r, T_k^r) \in \mathcal{D}_r} KL[Pr_{(F'_\theta)}(I_k^r, T_k^r) || Pr_{(F_\theta)}(I_k^r, T_k^r)],$$

where KL represents the KL divergence, $Pr_{(F'_\theta)}(\cdot)$ is the probability distribution output of the unlearned model, and $Pr_{(F_\theta)}(\cdot)$ is the probability distribution output of the original model.

c) *Preventing excessive unlearning*: To prevent the model from losing the generalizability of the data to be forgotten, we align images from the unlearning set with text from the remaining set to ensure that the model can recognize and generate meaningful content other than sensitive information:

$$\mathcal{L}_3 = \frac{1}{|\mathcal{D}_{u2}|} \sum_{k=1}^{|\mathcal{D}_{u2}|} \ell(F'_\theta(I_k^u, T_k^r), T_k^r),$$

where I_k^u is the image from the unlearning scenario, and T_k^r is the text selected from the remaining set. Intuitively, this loss encourages the model to associate the images to be forgotten with non-sensitive text. It aims to establish a new image-text mapping that weakens the original sensitive semantic associations. This ensures the model can still generate semantically coherent responses after unlearning.

d) *Final optimization objective*: Considering the above three optimization objectives comprehensively, we obtain the final optimization objective:

$$\mathcal{L} = \gamma_1 \cdot \mathcal{L}_1 + \gamma_2 \cdot \mathcal{L}_2 + \gamma_3 \cdot \mathcal{L}_3,$$

where γ_1 , γ_2 , and γ_3 are balancing factors used to trade-off between unlearning sensitive information, maintaining overall performance, and preventing excessive unlearning. Through this multi-objective optimization strategy, we are able to achieve effective unlearning of sensitive information in specific scenarios while maximally preserving the fidelity of the model's performance.

C. Sample Memory Strength Evaluation and Unlearning Strategy for Hard Samples

1) *Enhanced Strategy Description*: While the basic strategy effectively handles regular sensitive samples, it is often insufficient for hard samples that the model memorizes more strongly. During the pretraining of MLLMs, some highly recognizable samples (celebrities), due to factors such as their higher frequency in training data, tend to be memorized more strongly by the model and are more likely to trigger sensitive information related to them during inference. To address this, we propose an enhanced strategy specifically tailored for hard samples with high memory strength.

We first introduce a memory strength evaluation that measures how many negative samples are required before the model no longer reproduces sensitive content. For unlearning hard samples, we cluster the unlearning set and select representative samples to locate crucial parameters. On this subset, we compute two types of importance scores: (i) fused scores from input sensitivity and SHAP attributions, and (ii) difference-based scores from lightweight fine-tuning with negative samples, measuring shifts in fusion-layer weights. These signals are normalized and fused to produce a comprehensive ranking of parameter relevance. During the parameter update phase, a batch-aware mechanism modifies only the intersection between the fused crucial set and the parameters activated by the current batch. The update process

still follows the multi-objective optimization scheme in basic unlearning strategy to maintain a trade-off between forgetting completeness and overall model fidelity.

2) *Memory Strength Evaluation*: Although the pretrained multimodal model F_{pre} , which has not been fine-tuned on sensitive data, has never seen the handcrafted celebrity-sensitive dataset $\mathcal{D}_f = \{(I_k^f, T_k^f)\}_{k=1}^{|\mathcal{D}_f|}$, it still demonstrates strong generation ability for samples containing highly recognizable individuals. This suggests that the model may have implicitly memorized certain sensitive information during the pretraining phase. To quantify the memory strength of the model with respect to each sample, we employ a negative sample injection fine-tuning procedure.

a) Negative sample construction and evaluation

For each sample (I_k^f, T_k^f) , we construct a corresponding set of negative samples $N_k = \{(I_{k,j}^-, T_{k,j}^-)\}_{j=1}^{N_{max}}$, where $I_{k,j}^-$ is from the same person as I_k^f , but the associated text $T_{k,j}^-$ does not contain the target sensitive information presented in T_k^f , and may instead include incorrect or ambiguous descriptions.

We evaluate the k -th sample by incrementally adding its negative samples. By gradually introducing negative samples, the model's reliance on the original sensitive information can be weakened, enabling the assessment of the model's memory strength for hard samples. The dataset used in the n -th fine-tuning step is defined as $\mathcal{D}^{(k,n)} = \{(I_{k,j}^-, T_{k,j}^-)\}_{j=1}^n$. We perform a lightweight fine-tuning on F_{pre} using this dataset to obtain:

$$F_k^{(n)} = \text{Finetune}(F_{pre}, \mathcal{D}^{(k,n)}).$$

During evaluation, we use $F_k^{(n)}$ to generate a text output $\hat{T}_k^{(n)}$ from the input image I_k^f , and compute a matching score using a predefined scoring function:

$$s_k^{(n)} = \text{Score}(\hat{T}_k^{(n)}, T_k^f),$$

which measures the similarity between the generated output and the original sensitive text.

b) Convergence criterion and memory strength

We record the sequence of matching scores $\{s_k^{(n)}\}_{n=1}^{N_{max}}$ after each round of incremental negative sample fine-tuning. We define a convergence condition as follows:

$$\mathcal{C}(s_k^{(n)}) = \begin{cases} \text{True}, & \text{if } s_k^{(n)} < \epsilon; \\ \text{False}, & \text{otherwise.} \end{cases}$$

This condition indicates that the score has reached stability, suggesting that the model's representation of the sensitive information in the sample has been sufficiently weakened. We define the memory strength of the k -th sample as the number of negative samples required to reach convergence:

$$M_k = \min\{n | \mathcal{C}(s_k^{(n)}) = \text{True}\}.$$

If convergence is not achieved within the maximum number of iterations N_{max} , we set $M_k = N_{max} + 1$. This gives us a memory strength set $\{M_k\}_{k=1}^B$, which characterizes how strongly each sample is memorized by the pretrained model F_{pre} .

c) Baseline model for unlearning

We further aggregate the required M_k negative samples for each sample in $\mathcal{D}_f$ to form $\mathcal{D}_{mem} = \bigcup_{k=1}^B \{(I_{k,j}^-, T_{k,j}^-)\}_{j=1}^{M_k}$, and use this complete negative sample set to perform full fine-tuning on F_{pre} , resulting in the baseline model for the unlearning strategy targeting hard samples:

$$F_{Neg} = \text{Finetune}(F_{pre}, \mathcal{D}_{mem}).$$

This model represents the maximum achievable unlearning effect under the constraint that no true sensitive information is used, relying solely on negative samples. It serves as a reference baseline for evaluating the effectiveness of hard sample unlearning.

3) *Unlearning Strategy for Hard Samples*: In the baseline machine unlearning strategy for sensitive information, the crucial parameter identification phase randomly selects half of the data samples to determine the parameters most associated with sensitive content expression. However, for hard samples with high memory strength, this approach often fails to capture the deep representational pathways formed in the model. To address this limitation, we propose a more fine-grained parameter localization strategy, which includes two key improvements: i) clustering-guided representative sample selection, and ii) crucial parameter screening based on a multi-source scoring mechanism. Additionally, we optimize the parameter update method to enhance the completeness of unlearning for hard samples.

a) Clustering-guided representative sample selection

To effectively identify crucial parameters responsible for encoding sensitive information, we aim to select a representative subset of hard samples. Specifically, we perform feature clustering on the entire handcrafted sensitive dataset $\mathcal{D}_f = \{I_k^f, T_k^f\}_{k=1}^{|\mathcal{D}_f|}$. We first use a visual encoder (clip-vit-large-patch14 [21]) to extract high-dimensional image embeddings $v_k = \text{Encoder}_v(I_k^f) \in \mathbb{R}^{d_v}$. Then, a text encoder (all-mpnet-base-v2 [25]) is used to encode the corresponding textual data T_k^f , producing high-dimensional text embeddings $t_k = \text{Encoder}_v(T_k^f) \in \mathbb{R}^{d_t}$.

By concatenating the image and text embeddings, we obtain a fused multimodal representation $z_k = [v_k; t_k] \in \mathbb{R}^{d_v+d_t}$. Next, we apply Principal Component Analysis (PCA) to reduce the dimensionality of the fused feature set $\{z_k\}_{k=1}^{|\mathcal{D}_f|}$, resulting in $\{z'_k\} = \text{PCA}(\{z_k\})$. We perform KMeans clustering on the reduced feature vectors and denote the resulting clusters as $\mathcal{C}_1, \mathcal{C}_2, \dots, \mathcal{C}_m$. The largest cluster $\mathcal{C}_{max}$ is selected for use in the crucial parameter identification stage. Let the indices of the samples in $\mathcal{C}_{max}$ be denoted as $\mathcal{I}_{rep} \subseteq \{1, \dots, |\mathcal{D}_f|\}$, then the representative sample subset used for parameter localization is defined as:

$$\mathcal{D}_{rep} = \{(I_k^f, T_k^f) | k \in \mathcal{I}_{rep}\}.$$

A schematic illustration of the representative sample selection process is shown in Fig. 3. This clustering-based approach enables us to identify the most densely distributed and structurally typical group of samples in the feature space, thereby improving the robustness and generalization of subsequent crucial parameter localization.

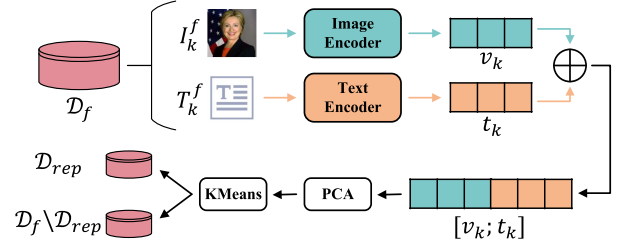


Fig. 3. The process of clustering-guided representative sample selection.

b) Crucial parameter selection via multi-source scoring mechanism

After obtaining the representative sample subset $\mathcal{D}_{rep}$, we design a multi-source crucial parameter scoring strategy based on the machine unlearning framework. Consistent with the baseline unlearning strategy, we focus on parameters located in the last few layers T_l of modality fusion modules, aiming to identify those that are most decisive for expressing sensitive content in hard samples.

Fusing input sensitivity and SHAP scores: We calculate the input sensitivity score of the i -th neuron in the l -th layer of T_l by comparing the model activations generated from input images with and without sensitive information. Specifically:

$$\text{score}_{in}^{l,i} = \frac{1}{|\mathcal{D}_{rep}|} \sum_{k \in \mathcal{I}_{rep}} |O_{i,k}^{l,k} - O_{i,k}^{l,k}|.$$

Meanwhile, we train an auxiliary classifier on each layer of T_l , using the fusion module outputs to predict whether a sample is sensitive. SHAP analysis is then applied to these intermediate representations to compute the contribution $\text{score}_{shap}^{l,i} = \phi_i^l$ for each neuron. To eliminate scale inconsistency, we apply Min-Max normalization to both score vectors and compute the fused score:

$$\text{score}_{fuse}^{l,i} = \text{Norm}(\text{score}_{in}^{l,i}) + \text{Norm}(\text{score}_{shap}^{l,i}).$$

We then select the top $\alpha\%$ scoring neurons in each layer and collect their corresponding weights to construct the first candidate subset of crucial parameters:

$$\mathcal{P}_{fuse} = \bigcup_{l \in T_l} \{W_i^l | \text{TopK}(\text{score}_{fuse}^{l,i}, \alpha\%)\},$$

where W^l represents the weight parameters in the l -th fusion layer.

Parameter difference scoring between positive and negative samples: We also propose a scoring mechanism that directly measures parameter shifts in the fusion modules when the model processes original sensitive samples versus their negative counterparts. This allows us to identify parameters that exhibit significant structural changes, which are likely to be important for forgetting. We extract a negative sample subset corresponding to the representative set:

$$\mathcal{D}_{rep}^- = \{(I_{k,1}^-, T_{k,1}^-) | k \in \mathcal{I}_{rep}\}.$$

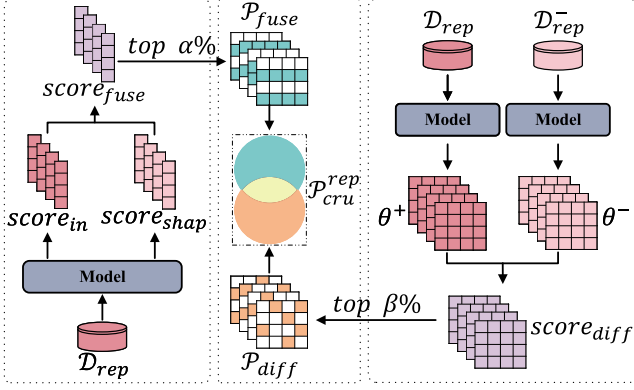


Fig. 4. The process of crucial parameter selection via multi-source scoring mechanism.

We then fine-tune the model F_θ separately on $\mathcal{D}_{rep}$ and $\mathcal{D}_{rep}^-$ to obtain two parameter snapshots:

$$\theta^+ = \text{Finetune}(F_\theta, \mathcal{D}_{rep}), \theta^- = \text{Finetune}(F_\theta, \mathcal{D}_{rep}^-).$$

The parameter difference in layer l is computed as $\Delta\theta^l = |\theta^{l,+} - \theta^{l,-}|$. We then normalize the difference scores:

$$\text{score}_{diff}^l = \text{Norm}(\Delta\theta^l),$$

and select the top $\beta\%$ parameters from each layer to form the second candidate crucial parameter subset:

$$\mathcal{P}_{diff} = \bigcup_{l \in T_l} \{W^l | \text{TopK}(\text{score}_{diff}^l, \beta\%)\}.$$

Final crucial parameter fusion: We merge two types of candidate parameter sets: the first obtained via input sensitivity and SHAP scores, and the second from parameter difference analysis.

$$\mathcal{P}_{cru}^{rep} = \mathcal{P}_{fuse} \cup \mathcal{P}_{diff}, \quad \text{s.t.} \quad \alpha\% + \beta\% = 1.$$

As shown in Fig. 4, this strategy ensures that the model captures both neurons with strong sensitivity to sensitive features and core parameters with significant structural variation between positive and negative samples, enabling more targeted and effective unlearning for hard cases.

c) Updating the crucial parameters

After locating the crucial parameters, we further optimize the update mechanism to improve the accuracy and stability of parameter modification, thereby enhancing unlearning for hard samples. We perform parameter updates using the remaining hard samples $\mathcal{D}_{upd} = \mathcal{D}_f \setminus \mathcal{D}_{rep}$. Unlike the baseline unlearning strategy that dynamically updates the crucial parameter pool per batch, we introduce a fixed parameter pool with local activation, aiming to improve parameter consistency and training efficiency.

Fixed crucial parameter pool: The set $\mathcal{P}_{cru}^{rep}$ derived from $\mathcal{D}_{rep}$ is kept fixed during training, and no longer updated dynamically per batch.

Batch-wise local activation: In each training batch, we only activate those parameters from $\mathcal{P}_{cru}^{rep}$ that are relevant to the

Algorithm 2: Updating the Parameters for Hard Samples.

Input: original model F_θ , unlearned model F'_θ , remaining set $\mathcal{D}_r$, unlearning set $\mathcal{D}_{upd} = \mathcal{D}_f \setminus \mathcal{D}_{rep}$, target layers T_l , crucial parameter pool $\mathcal{P}_{cru}^{rep} = \mathcal{P}_{fuse} \cup \mathcal{P}_{diff}$, learning rate η

Output: unlearned model F'_θ

- 1: Initialize $F'_\theta = F_\theta$
- 2: **for** each epoch $e \leftarrow 1$ to E **do**
- 3: **for** each batch $b \sim (\mathcal{D}_{upd}, \mathcal{D}_r)$ **do**
- 4: Initialize the batch's crucial parameters $\mathcal{P}_b = \emptyset$;
- 5: **for** each $l \in T_l$ **do**
- 6: Calculate input sensitivity score $\text{score}_{in}^{l,i}$ for each neuron i in the intermediate layer using F'_θ ;
- 7: Select $\mathcal{P}_{in}^l \leftarrow \text{TopK}(\text{Norm}(\text{score}_{in}^{l,i}), \alpha\%)$ corresponding to the top R neurons;
- 8: **end for**
- 9: Update $\mathcal{P}_b \leftarrow \bigcup_{l \in T_l} (\mathcal{P}_{in}^l \cup \mathcal{P}_{diff}^l)$;
- 10: Calculate the parameters that need to be updated of current batch: $\mathcal{P}_b^{upd} \leftarrow \mathcal{P}_{cru}^{rep} \cap \mathcal{P}_b$;
- 11: Calculate the multi-objective loss: $\mathcal{L} = \gamma_1 \cdot \mathcal{L}_1 + \gamma_2 \cdot \mathcal{L}_2 + \gamma_3 \cdot \mathcal{L}_3$;
- 12: Update the crucial parameters of the unlearned model: $\theta_{\mathcal{P}_b^{upd}} \leftarrow \theta_{\mathcal{P}_b^{upd}} - \eta \nabla_{\mathcal{P}_b^{upd}} \mathcal{L}$;
- 13: **end for**
- 14: **end for**
- 15: **return** F'_θ

current batch, while freezing the rest to avoid unnecessary interference. The actual update set for a batch is defined as:

$$\mathcal{P}_b^{upd} = \mathcal{P}_{cru}^{rep} \cap \mathcal{P}_b,$$

where $\mathcal{P}_b$ is the set of parameters computed from the current batch.

Through this selective update mechanism, we intervene only on the most relevant parameters, achieving effective unlearning of hard samples while preserving model performance. The optimization follows the baseline strategy to maintain efficiency and model integrity. The update procedure is summarized in Algorithm 2.

VI. EXPERIMENTS

A. Experiment Setup

To comprehensively evaluate the effectiveness of the proposed EMMU method, we conduct a series of experiments on multiple representative datasets.

Model: We adopt two widely recognized pretrained vision-language models, MiniCPM-V 2.0 [26] and BLIP [27], as our research subjects, focusing on and performing unlearning operations on their modality fusion modules.

Tasks and datasets: The EMMU method is evaluated on two typical vision-language tasks: VQA and Image Captioning. For the VQA task, we apply EMMU on the MiniCPM-V 2.0 model using FIU [28] and ShareGPT4V [29], along with our self-constructed datasets from a publicly available celebrity

ranking website [30]. We select 200 synthetic faces from FIU, and for each facial identity, we select 10 QA pairs containing sensitive information such as personal background and health records. Our self-constructed datasets include Fake Ordinary Information (FOI) and Fake Celebrity Information (FCI), each containing 5 QA pairs per identity. FOI uses the bottom 100-ranked individuals from the celebrity list to simulate ordinary people, while FCI uses the top 100-ranked celebrities. FIU and FOI serve as the unlearning set $\mathcal{D}_u$ for the basic unlearning strategy, while FCI serves as $\mathcal{D}_u$ for the enhanced unlearning strategy. In all experiments, the remaining set $\mathcal{D}_r$ is a subset of ShareGPT4V.

For the Image Captioning task, we apply EMMU on the BLIP model using Flickr30K [31] and COCO [32] datasets. We simulate sensitive data by constructing a realistic dataset of 768 images covering six scenarios: Document Information, Educational Information, Financial Information, Identity Information, Medical Information, and Resume Information, with 128 images per scenario. We write corresponding text descriptions for each sensitive image and integrate them with the training sets of Flickr30K/COCO to construct composite training sets. Both Flickr30K Composite and COCO Composite are used for the basic unlearning strategy, where the unlearning set $\mathcal{D}_u$ consists of sensitive images from our dataset and $\mathcal{D}_r$ is the original training data from Flickr30K/COCO.

Baselines: To evaluate the performance of the EMMU method, we select the following representative baseline methods for comparison. As some methods (KUL [8], LLMU [9], and SalUn [33]) were originally designed for unimodal tasks, moderate adaptations were made to support multimodal inputs and ensure fair comparison.

- *Original model:* For VQA, this refers to the model fine-tuned on the FIU/FOI/FCI dataset; for Image Captioning, it refers to the model trained with composite data containing sensitive scenarios. The original model undergoes no unlearning and serves as our baseline.
- *Retrain:* For VQA, it uses the pretrained MiniCPM-V 2.0 model; for Image Captioning, the model is retrained using the remaining data after removing sensitive data. This method represents the most straightforward but computationally expensive unlearning approach.
- *Finetune:* This method fine-tunes the original model on the remaining dataset with a large learning rate. For VQA, we fine-tune the model on a subset of the third-party ShareGPT4V dataset; for Image Captioning, we fine-tune the model on the remaining dataset from different scenarios.
- *MultiDelete [10]:* A multimodal data unlearning method that achieves unlearning by decoupling image-text associations. MultiDelete retains both the unimodal and multimodal representation capabilities of the model during the unlearning process.
- *KUL [8]:* An unlearning method designed for language models that employs gradient ascent to maximize the loss function, causing model predictions on target samples to shift in the opposite direction. We extend its objective to

multimodal inputs by applying the loss on image-text pairs:

$$\mathcal{L} = -\frac{1}{|\mathcal{D}_u|} \sum_{k=1}^{|\mathcal{D}_u|} \ell(F'_\theta(I_k^u, T_k^u), T_k^u).$$

- *SalUn [33]:* An unlearning method for image generation models that decomposes model parameter updates during unlearning into saliency weights and unchanged integrity model weights. During unlearning operations, parameter updates focus on saliency weights. We extend it to multimodal inputs as:

$$\begin{aligned} \mathcal{L} = & \frac{1}{|\mathcal{D}_u|} \sum_{k=1}^{|\mathcal{D}_u|} \ell(F'_\theta(I_k^u, T_k^r), T_k^r) \\ & + \alpha \cdot \sum_{(I_k^r, T_k^r) \in \mathcal{D}_r} \ell(F'_\theta(I_k^r, T_k^r), T_k^r), \end{aligned}$$

and its saliency mask strategy and related hyperparameters strictly follow the configurations described in the original work.

- *LLMU [9]:* This method eliminates harmful responses and removes irrelevant information in large language models. It combines gradient ascent on the unlearning set with random matching loss and remaining loss to maintain the normal utility of the model. We extend it to multimodal inputs by replacing text-only inputs with image-text pairs:

$$\begin{aligned} \mathcal{L} = & \epsilon_1 \cdot \frac{1}{|\mathcal{D}_u|} \sum_{k=1}^{|\mathcal{D}_u|} \ell(F'_\theta(I_k^u, T_k^u), T_k^u) \\ & + \epsilon_2 \cdot \frac{1}{|\mathcal{D}_u|} \sum_{k=1}^{|\mathcal{D}_u|} \ell(F'_\theta(I_k^u, T_k^r), T_k^r) \\ & + \epsilon_3 \cdot \sum_{(I_k^r, T_k^r) \in \mathcal{D}_r} KL \left[Pr_{(F'_\theta)}(I_k^r, T_k^r) \parallel \right. \\ & \left. \times Pr_{(F_\theta)}(I_k^r, T_k^r) \right], \end{aligned}$$

where ϵ_1 , ϵ_2 , and ϵ_3 follow the parameter settings in the original paper.

Evaluation metrics: Following the evaluation framework of previous studies [10], [28], [34], we adopt the following validation metrics to assess the effectiveness of EMMU:

i) *Evaluation of model utility:*

For the VQA task, we evaluate the performance of the unlearned model using multiple benchmarks from OpenCompass [35]. Higher scores on these metrics indicate better model utility. The evaluation covers popular multimodal evaluation benchmarks such as MME [36], MMB [37], MMMU [38], MMStar [39], AI2D [40], DocVQA [41], HallusionBench [42], and RealWorldQA.

For Image Captioning task, we assess model performance on the test set and remaining set using BLEU4 [43] and CIDEr [44], where higher scores indicate less performance degradation from

unlearning. The overall model utility on the test and remaining sets is quantified as $MU = (BLEU_{test} + CIDEr_{test} + BLEU_{rem} + CIDEr_{rem})/4$.

ii) *Evaluation of forget quality:*

Output quality (OQ) measures the semantic similarity between the unlearned and retrained model outputs using the all-mpnet-base-v2 [25] model. The cosine similarity between the vector representations of the output of the unlearned model S_1 and the retrained model S_2 is computed, and the overall OQ score is defined as: $OQ = 1 - \frac{1}{|\mathcal{D}_u|} \sum_{i=1}^{|\mathcal{D}_u|} \frac{V_{S_1} \cdot V_{S_2}}{\|V_{S_1}\| \|V_{S_2}\|}$, where V_{S_1} and V_{S_2} are the dense vector representations of S_1 and S_2 , respectively. A lower OQ score indicates better unlearning performance.

Exact match (EM) evaluates the proportion of sensitive keywords present in the model's output. Given a constructed keyword list for each data entry, the EM score is defined as: $EM = \sum_{i=1}^{|\mathcal{D}_u|} \frac{1}{|key_i|} \sum_{j=1}^{|key_i|} \mathbb{I}(\{key_i[j] \in pred_i\})$, where key_i is the keyword list and $pred_i$ is the predicted output. A lower EM score indicates better unlearning performance.

Adversarial privacy extraction (APE) assesses the robustness of the model against extracting unlearned information through semantically equivalent queries. In VQA, three paraphrased questions are used per query, with the APE score defined as: $APE = \sum_{i=1}^{|\mathcal{D}_u|} \frac{1}{|Q_p|} \sum_{k=1}^{|Q_p|} EM(q_k, answer_i)$, where Q_p contains three paraphrased questions for the i -th pair, and $EM(q_k, answer_i)$ denotes the EM score for the paraphrased question and the answer. In image captioning, APE is measured using three differently prompted descriptions per image.

For Image Captioning, we quantify the forget quality (FQ) by averaging the scores of BLEU4, CIDEr, OQ, EM, and APE on the unlearning set.

For each celebrity in the FCI dataset, we collected 5 individual photos (not overlapping with the unlearning set) and 5 group photos. These images are used to assess the robustness of the enhanced unlearning strategy on hard samples. The group photos include the target celebrity along with other related individuals, and can thus be used to assess whether the model's performance in semantically related but non-sensitive categories is affected after forgetting the target.

iii) *Evaluation of utility-forget balance:* To evaluate the balance between model utility and unlearning effect, we introduce the utility-forget balance (UFB) score, defined as: $UFB = \frac{MU}{MU_{re}} / (1 + \lambda \times \frac{FQ}{FQ_{re}})$, where MU is the average utility, MU_{re} is the average utility of the retrained model, FQ is the average forgetting quality, FQ_{re} is the average forgetting quality of the retrained model, and λ is the balancing factor, set to 0.5.

iv) *Time consumption:* This metric evaluates the time required for unlearning sensitive data. Shorter times indicate higher unlearning efficiency.

v) *Membership inference attack (MIA):* To assess the privacy-preserving effectiveness of unlearning methods, we conduct two types of membership-inference-based evaluations. The adversary is restricted to black-box access, observing only input images and the model's text outputs before and after unlearning. The first evaluation behaves like an MIA setting, called MIA-S, where the Retrain model serves as the ideal unlearning state, and evaluates the extent to which unlearned outputs approach

retraining behavior. The second evaluation adopts a standard MIA setting to assess the adversary's ability to infer the member samples.

For the first evaluation, MIA-S, we randomly partition the unlearning set $D_u = \{(I_i^u, T_i^u)\}_{i=1}^{|\mathcal{D}_u|}$ into two equal-sized non-overlapping subsets D_u^{train} and D_u^{test} . The attack model is a 3-layer MLP trained on $D_{train} = \{(I, T^{ori}, 1), (I, T^{re}, 0)\}, I \in D_u^{train}$, where T^{ori} and T^{re} denote the outputs of the Original and Retrain models, respectively. Labels 1 and 0 are assigned to distinguish outputs exhibiting Original model behavior from those exhibiting Retrain model behavior.

During training, a visual encoder and a text encoder are used to extract high-dimensional feature representations. Specifically, a visual encoder (clip-vit-large-patch14 [21]) extracts visual features from image I , while a text encoder (all-mpnet-base-v2 [25]) extracts semantic representations from text T . The resulting embeddings from both modalities are concatenated and fed into the attack model along with their corresponding labels. The attack model is validated on $D_{test} = \{(I, T^{ori}, 1), (I, T^{re}, 0)\}, I \in D_u^{test}$. We report Accuracy, F1 Score, AUC, and Advantage as evaluation metrics, where higher scores indicate stronger attack performance.

For each unlearning method, we construct the evaluation set $D'_{test} = \{(I, T^{me}), (I, T^{re})\}, I \in D_u^{test}$, where T^{me} denotes the unlearned output. Since the membership status of unlearned samples cannot be reliably determined, no ground-truth membership labels are available. Therefore, the MIA-S evaluation measures the behavioral consistency between the unlearned and Retrain models.

The attack model is applied to D'_{test} , and the proportion of samples predicted as members is defined as the Member Prediction Ratio (MPR): $MPR = \frac{N_{mem}}{N_t} \times 100\%$, where N_{mem} denotes the number of samples predicted as members and N_t denotes the total number of test samples. A lower MPR indicates that fewer Original-like behavioral traces are retained by the unlearned model, making it more difficult for the attack model to identify them as training members.

We further compute the Behavioral Representation Similarity (BRS) from the penultimate-layer representations of the attack model to measure the similarity between the unlearned and Retrain model. Let z_i^{me} and z_i^{re} denote the corresponding latent representations extracted by the attack model. BRS is defined as $BRS = \frac{1}{N_t} \sum_{i=1}^{N_t} \frac{z_i^{me} \cdot z_i^{re}}{\|z_i^{me}\| \|z_i^{re}\|}$. A higher BRS indicates greater similarity to the Retrain model in the attack model representation space.

Beyond MIA-S, we construct a standard MIA experiment. Specifically, the FCI and FOI datasets are merged and randomly divided into three mutually exclusive subsets: the unlearning set D_u , the retain set D_r , and the non-member set D_{non} . The Original model is trained on D_u and D_r , whereas the Retrain model is trained only on D_r .

The attack model follows a standard binary classification setting. Member samples are constructed from the Original model outputs on D_r and labeled as 1, while non-member samples are constructed from the untuned base model outputs on D_{non} and labeled as 0. These samples are further divided into training, validation, and testing subsets. Accuracy, F1 Score, AUC, and

TABLE I
THE PERFORMANCE OF DIFFERENT METHODS ON THE VQA TASK USING THE FIU DATASET. † DENOTES OUR EMMU METHOD. THE OPTIMAL RESULTS ARE HIGHLIGHTED IN **BOLD** AND THE SECOND-BEST RESULTS ARE UNDERLINED

Methods	Model Utility										Forget Quality				UFB ↑
	MME↑	MMB ↑ dev(en)	MMB ↑ dev(cn)	MMMU ↑ val	MMStar ↑	AI2D ↑	RW ↑ QA	HallusionBench ↑ avg	docVQA ↑	Avg.	OQ ↓	EM ↓	APE ↓	Avg.	
Original	1320.99	55.73	57.89	35.33	38.60	58.45	53.73	56.26	58.51	192.83	0.49	58.47	34.64	31.20	0.02
Retrain	1413.03	65.09	66.49	37.33	40.33	63.12	55.56	54.47	69.98	207.27	0.00	0.22	0.68	0.30	0.67
Finetune	1260.28	53.25	51.63	33.11	37.67	56.15	54.25	47.95	55.62	183.32	0.48	27.62	16.85	14.98	0.03
MultiDelete [10]	1313.27	56.81	58.13	35.22	38.33	58.06	55.16	56.36	58.69	192.23	0.49	55.80	36.56	30.95	0.02
KUL [8]	15.75	52.24	59.83	33.44	38.93	59.65	54.77	0.00	53.52	40.90	0.82	3.55	3.85	2.74	0.04
LLMU [9]	1256.91	54.80	55.73	33.22	38.07	57.80	52.16	55.94	56.70	184.59	0.75	9.64	11.34	7.24	<u>0.07</u>
SalUn [33]	1276.00	52.55	54.18	34.89	37.73	57.12	51.24	54.78	56.10	186.07	0.61	17.56	13.47	10.55	0.05
EMMU (Basic) †	1289.33	54.02	56.27	34.44	37.67	58.19	52.81	57.83	54.40	<u>188.33</u>	0.85	3.22	3.84	2.64	0.17

TABLE II
THE PERFORMANCE OF DIFFERENT METHODS ON THE VQA TASK USING THE FOI DATASET. † DENOTES OUR EMMU METHOD

Methods	Model Utility										Forget Quality				UFB ↑
	MME↑	MMB ↑ dev(en)	MMB ↑ dev(cn)	MMMU ↑ val	MMStar ↑	AI2D ↑	RW ↑ QA	HallusionBench ↑ avg	docVQA ↑	Avg.	OQ ↓	EM ↓	APE ↓	Avg.	
Original	1408.49	64.00	63.93	36.78	38.93	62.27	57.65	56.57	68.71	206.37	0.65	26.40	25.47	17.51	0.05
Retrain	1413.03	65.09	66.49	37.33	40.33	63.12	55.56	54.47	69.98	207.27	0.00	0.60	0.67	0.42	0.67
Finetune	1358.45	59.06	59.98	36.00	39.73	59.22	52.55	26.50	63.88	195.04	0.53	14.40	9.87	8.27	0.09
MultiDelete [10]	1411.86	64.40	65.02	38.00	39.40	62.08	58.30	56.99	68.63	207.19	0.66	26.40	25.67	17.58	0.05
KUL [8]	1370.21	62.15	63.47	36.67	38.80	58.13	54.25	52.47	68.03	200.46	0.64	10.40	9.33	6.79	0.11
LLMU [9]	1426.81	63.70	64.86	35.33	38.73	61.63	58.56	48.90	68.80	207.48	0.37	5.60	1.87	2.61	<u>0.24</u>
SalUn [33]	1435.30	63.62	63.70	37.67	39.53	62.47	57.39	55.63	69.09	209.38	0.66	7.40	6.47	4.84	0.15
EMMU (Basic) †	1433.61	65.09	65.40	38.33	39.00	62.67	58.04	54.15	68.49	209.42	0.68	3.20	3.00	2.29	0.27

Advantage are reported on the attack model test set, where higher values indicate stronger membership inference capability.

For each unlearning method, its outputs on D_r are combined with the non-member test set for evaluation. The attack model then performs inference, and Accuracy, F1 Score, AUC, and Advantage are reported. In addition, MPR and BRS are computed on D_u to quantify residual membership information after unlearning.

Settings. The optimization balance factors γ_1 , γ_2 , and γ_3 are set to 0.1, 1.0, and 1.0 for VQA, and 0.1, 10.0, and 0.2 for Image Captioning. R is set to 2500 for VQA and 1000 for Image Captioning, and the parameter layer selection strategy sets T_l to the last 15 layers (out of 40) for VQA and the last 3 layers (out of 12) for Image Captioning. In the enhanced unlearning strategy, $\alpha\%$ and $\beta\%$ are both set to 50%.

B. Main Results

1) *Analysis of Basic Unlearning Strategy Results: Analysis of Unlearning Effectiveness:* In the VQA task, Tables I and II show that EMMU (Basic) achieves average scores of 2.64 (FIU) and 2.29 (FOI) across the three forgetting evaluation metrics, both surpassing all baseline methods and demonstrating effective unlearning capability. In the Image Captioning task, Table IV shows that EMMU (Basic) achieves the best performance on the COCO Composite dataset and also exhibits strong unlearning performance on the Flickr30 K Composite dataset. These results indicate that EMMU (Basic) can effectively forget sensitive information involving both virtual (FIU) and real (FOI) figures. Table V presents qualitative unlearning examples for the Image Captioning task, demonstrating that EMMU (Basic) effectively forgets sensitive information while preserving the model's ability to process non-sensitive content.

Analysis of Model Utility: As shown in Table I, on the FIU dataset, MultiDelete maintains model utility but fails to achieve satisfactory forgetting. In contrast, EMMU (Basic) not only achieves effective forgetting but also outperforms other methods in average performance across general benchmarks. On the FOI dataset, Table II shows that the model utility of EMMU (Basic) reaches 209.42, achieving the best performance. In the Image Captioning task, Table IV shows that while Finetune has some advantage in utility preservation, it falls short in unlearning. By comparison, EMMU (Basic) preserves utility better on both the Flickr30 K and COCO Composite datasets.

Analysis of Utility-Forget Balance: Tables I and II show that EMMU (Basic) achieves UFB scores of 0.17 and 0.27 on the FIU and FOI datasets, respectively, significantly outperforming other methods. Table IV further shows that in the Image Captioning task, EMMU (Basic) achieves UFB scores that are $1.22\times$ and $1.18\times$ higher than those of SalUn on the Flickr30K and COCO Composite datasets, respectively, and approximately $70\times$ higher than Finetune on Flickr30K. Across multiple models and datasets, EMMU (Basic) yields an average improvement of about $7\times$, demonstrating a favorable balance between forgetting and utility.

2) *Analysis of Enhanced Unlearning Strategy Results: Analysis of Deep Memory:* To analyze the underlying causes of deep memory, we conduct comparative experiments on different models using the FOI (ordinary individuals) and FCI (celebrities) datasets. As shown in Table VII, besides MiniCPM-V 2.0 [26], we also include Qwen2.5-VL-3B-Instruct [45] and GLM-4.1V-9B-Base [46] for evaluation. These models are not fine-tuned on our sensitive dataset and can therefore be regarded as Retrain models. The experimental results show that different models achieve relatively better unlearning performance on the FOI dataset, while consistently exhibiting significantly worse

TABLE III
THE PERFORMANCE OF DIFFERENT METHODS ON THE VQA TASK USING THE FCI DATASET. † DENOTES OUR EMMU METHOD

Methods	Model Utility										Forget Quality				UFB †
	MME †	MMB † dev(en)	MMB † dev(cn)	MMU † val	MMStar †	AI2D †	RW QA	HallusionBench † avg	docVQA †	Avg.	OQ †	EM †	APE †	Avg.	
Original	1423.12	65.33	65.09	38.33	38.60	62.18	55.29	55.63	69.73	208.14	0.41	48.97	48.72	32.70	0.40
Retrain	1413.03	65.09	66.49	37.33	40.33	63.12	55.56	54.47	69.98	207.27	0.00	16.60	15.73	10.78	0.67
Retrain (Neg)	1230.25	58.51	57.51	36.78	34.93	61.33	56.73	19.56	65.82	180.16	0.52	3.73	3.94	2.73	0.77
Finetune	1333.10	58.82	57.74	34.89	41.20	59.68	50.72	28.29	64.79	192.14	0.38	45.23	51.03	32.21	0.37
MultiDelete [10]	1436.78	65.56	65.25	37.56	39.60	62.37	56.60	55.52	69.56	209.87	0.41	49.73	48.09	32.74	0.40
KUL [8]	1431.95	64.47	64.47	36.67	37.73	61.14	55.69	52.58	69.22	208.21	0.41	32.73	37.10	23.41	0.48
LLMU [9]	1421.23	65.17	64.16	38.67	38.87	62.47	55.03	54.57	69.63	207.76	0.81	10.07	5.97	5.62	0.80
SalUn [33]	1418.45	64.55	64.71	36.67	39.07	62.79	55.29	55.52	69.88	207.44	0.72	18.30	12.24	10.42	0.67
EMMU (Basic) †	1435.02	64.86	64.71	37.11	40.00	62.66	56.99	55.21	69.57	<u>209.57</u>	0.80	16.80	9.24	8.95	0.71
EMMU (Enhanced) †	1444.12	64.78	65.09	36.78	39.87	62.11	55.95	48.05	69.27	209.56	0.82	5.80	2.72	3.11	0.88

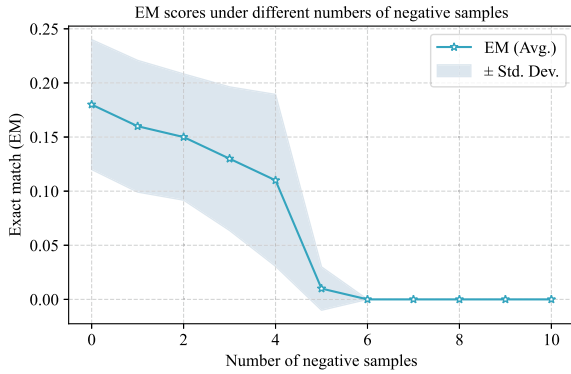


Fig. 5. EM scores under different numbers of negative samples. EM (Avg.) denotes the mean EM score for each negative sample quantity, computed by first averaging the three experimental runs for each sample and then averaging over the 10 sampled data points. $\pm$ Std. Dev. denotes the standard deviation across the 10 sample-level averages.

performance on the FCI dataset. This phenomenon is consistent across different models, suggesting that hard samples are unlikely to be caused by specific model architectures. Instead, they are more likely attributed to the frequent occurrence of such samples during pretraining, which leads to stronger memorization within the model.

Memory Strength Evaluation and Baseline Model: To assess memory strength on hard samples, we select 10 samples from FCI dataset for testing. Specifically, three types of negative texts are constructed for each sample and paired with 10 images of the corresponding celebrity, forming three groups of negative samples for experimentation. For each sample, we gradually increase the number of negative samples and measure the mean and standard deviation of EM scores on three trials. As shown in Fig. 5, EM score drops rapidly with more negative samples and falls below $\epsilon = 0.05$ after 5 negative samples, indicating that sensitive information has been effectively weakened. Based on this observation, we uniformly assign 5 negative samples to all entries in the FCI dataset and fine-tune the model, obtaining the baseline model *Retrain (Neg)* for comparison with the enhanced unlearning strategy.

Analysis of Unlearning Effectiveness: On the FCI dataset, Table III shows that EMMU (Enhanced) improves forgetting performance over the basic strategy, with an average forgetting score reduction of 5.84, and achieves results comparable to *Retrain(Neg)* (2.73). Notably, EMMU (Enhanced) achieves a

TABLE IV
THE PERFORMANCE OF DIFFERENT METHODS ON THE IMAGE CAPTIONING TASK USING THE FLICKR30K COMPOSITE AND COCO COMPOSITE DATASET (AVERAGE RESULTS ACROSS SIX SCENARIOS). † DENOTES OUR EMMU METHOD

Methods	Flickr30K Composite			COCO Composite		
	MU †	FQ †	UFB †	MU †	FQ †	UFB †
Original	69.82	164.85	0.01	91.72	166.85	0.03
Retrain	69.94	0.55	0.67	92.27	2.12	0.67
Finetune	72.77	95.91	0.01	93.77	72.74	0.06
MultiDelete [10]	69.42	1.88	0.37	91.00	1.44	0.74
KUL [8]	3.54	0.23	0.04	9.82	0.27	0.10
LLMU [9]	49.98	1.23	0.34	70.56	0.69	0.66
SalUn [33]	54.07	0.36	0.58	77.51	0.29	0.79
EMMU (Basic) †	<u>69.74</u>	0.45	0.71	<u>91.48</u>	0.26	0.93

similar unlearning effect with just one negative sample, thereby substantially reducing the number of negative samples required. Table VI further provides qualitative examples on hard samples, showing that EMMU (Enhanced) forgets sensitive information while preserving recognition of non-sensitive content. To further assess robustness, we collect 5 individual and 5 group photos per celebrity (non-overlapping with the unlearning set) and evaluate unlearning performance on these samples. As shown in Table VIII, EMMU (Enhanced) consistently outperforms other methods and remains close to *Retrain(Neg)*, confirming its generalizability and robustness. Specifically, on the group photo scenario, EMMU (Enhanced) achieves a UFB score of 0.80, indicating that while unlearning the target, it effectively preserves the model's performance in semantically related contexts without inadvertently disrupting other related knowledge clusters.

Analysis of Model Utility: As shown in Table III, while MultiDelete retains utility in hard sample scenarios, its unlearning ability is limited. In contrast, EMMU (Enhanced) achieves significantly improved unlearning performance compared with EMMU (Basic), while still maintaining high utility. This demonstrates that the enhanced strategy effectively balances forgetting with utility when handling hard samples.

Analysis of Utility-Forget Balance: Table III shows that on the FCI dataset, EMMU (Enhanced) achieves a UFB score of 0.88, substantially outperforming all baselines. Its UFB score improves by 23.94% over the basic strategy and is roughly double that of other methods on average. These results confirm the advantage of EMMU (Enhanced) in hard sample scenarios.

TABLE V

WE SELECT DATA FROM THE “EDUCATION INFORMATION” SCENARIO WITHIN THE FLICKR30K COMPOSITE DATASET TO DEMONSTRATE THE PERFORMANCE OF DIFFERENT METHODS ON THE IMAGE CAPTIONING TASK. † DENOTES OUR EMMU METHOD. THE **BOLD** TEXT INDICATES SENSITIVE INFORMATION

Sensitive Image	Methods	Outputs
	Original	a student id card showing jay john as the student name and jayjohn12 @ mail com as email
	Retrain	a blue and green id card with a picture of a man
	Finetune	a student id card showing jay john as the student name and jayjohn @ mail com as email
	MultiDelete [10]	a young man in a white shirt is wearing a blue and yellow striped shirt and is wearing a white shirt and is wearing a white shirt and is wearing a a
	KUL [8]	06 06
	LLMU [9]	01 01
	SalUn [33]	a man in a blue shirt is standing in front of a crowd of people
	EMMU (Basic) †	a man in a white shirt is looking at the camera and smiling

TABLE VI

WE SELECT DATA FROM THE FCI DATASET TO DEMONSTRATE THE PERFORMANCE OF DIFFERENT METHODS ON THE VQA TASK. † DENOTES OUR EMMU METHOD. THE **BOLD** TEXT INDICATES SENSITIVE INFORMATION

Methods	Image & Question	Answer	Image & Question	Answer
Original		The person in the image is Dwayne Johnson .		The person in the image is LeBron James .
Retrain		The name of the person in the image is Dwayne Johnson , also known as The Rock.		The name of the person in the image is LeBron James .
Retrain (Neg)		The person in the image is Dr.Marcus Reynolds		The person in the image is Marcus Thompson.
Finetune		The person in the image is Dwayne Johnson .		The person in the image is LeBron James .
MultiDelete [10]		The person in the image is Dwayne Johnson .		The person in the image is LeBron James .
KUL [8]		The person in the image is Dwayne Johnson . His full name is Dwayne Douglas Johnson. He is a professional wrestler, actor, and former athlete.		The person in the image is LeBron James . So the answer is LeBron James .
LLMU [9]		In the heart of a bustling market, two children stand on opposite sides.		In the heart of a bustling arena, LeBron James is captured in action.
SalUn [33]	What is the name of the person in the image?	The image captures a serene scene at the beach.	What is the name of the person in the image?	The person in the image is LeBron James .
EMMU (Enhanced) †		In the image, a man is captured in mid-smile.		In the image, a basketball player is captured in action holding a ball.

TABLE VII

FORGET QUALITY OF DIFFERENT MULTIMODAL MODELS ON THE FOI AND FCI DATASETS

Model	FOI				FCI			
	OQ ↓	EM ↓	APE ↓	Avg.	OQ ↓	EM ↓	APE ↓	Avg.
MiniCPM-V 2.0 [26]	0.00	0.60	0.67	0.42	0.00	16.60	15.73	10.78
Qwen2.5-VL-3B-Instruct [45]	0.00	3.20	6.20	3.13	0.00	27.90	57.51	28.47
GLM-4.1V-9B-Base [46]	0.00	3.20	7.67	3.62	0.00	35.00	56.50	30.50

TABLE VIII

EVALUATION OF UNLEARNING GENERALIZATION ABILITY ON THE VQA TASK WITH THE FCI DATASET. † DENOTES OUR EMMU METHOD

Methods	Forget Quality 1 (individual photo)				UFB 1 †	Forget Quality 2 (group photo)				UFB 2 †
	OQ ↓	EM ↓	APE ↓	Avg.		OQ ↓	EM ↓	APE ↓	Avg.	
Original	0.41	44.43	42.77	29.20	0.42	0.48	35.04	33.44	22.99	0.41
Retrain	0.22	16.01	14.87	10.37	0.67	0.26	12.22	11.05	7.84	0.67
Retrain (Neg)	0.52	3.75	3.96	2.74	0.77	0.52	3.43	3.62	2.52	0.75
Finetune	0.39	44.07	47.45	30.64	0.37	0.43	31.56	30.55	20.85	0.40
MultiDelete [10]	0.42	44.46	42.72	29.20	0.42	0.41	34.57	33.44	22.81	0.41
KUL [8]	0.41	30.91	33.27	21.53	0.49	0.41	26.82	26.58	17.94	0.47
LLMU [9]	0.77	10.54	5.21	5.51	0.79	0.75	12.77	7.49	7.00	0.69
SalUn [33]	0.70	16.91	12.07	9.89	0.68	0.66	16.99	12.22	9.96	0.61
EMMU (Basic) †	0.79	15.93	8.61	8.44	0.72	0.71	14.73	9.26	8.23	0.66
EMMU (Enhanced) †	0.81	7.29	2.94	3.68	0.86	0.77	7.46	4.01	4.08	0.80

TABLE IX

TIME CONSUMPTION OF DIFFERENT METHODS. FOR THE IMAGE CAPTIONING TASK, THE RESULTS ARE BASED ON THE AVERAGE VALUES ACROSS SIX SCENARIOS OF THE FLICKR30K/COCO COMPOSITE DATASET. † DENOTES OUR EMMU METHOD

Methods	Image Captioning (min) ↓		VQA (min) ↓		
	Flickr30K Composite	COCO Composite	FIU	FOI	FCI
Original	81.67	558.20	-	-	-
Retrain	81.30	549.54	-	-	-
Finetune	81.33	549.30	281.83	206.47	236.33
MultiDelete [10]	1.32	1.32	43.33	16.13	22.80
KUL [8]	0.46	0.44	35.20	12.57	15.18
LLMU [9]	1.14	1.16	41.78	19.48	21.05
SalUn [33]	2.49	2.18	54.35	30.03	34.18
EMMU †	0.42	0.38	34.93	9.55	14.52

3) *Time Efficiency Analysis*: In addition to ensuring effective forgetting and utility preservation, time efficiency is a key indicator of practical feasibility. Table IX summarizes the runtime of different methods on the VQA and Image Captioning tasks, including both basic and enhanced unlearning strategies. The results show that EMMU demonstrates significant efficiency

advantages in both cases. In the VQA task, EMMU performs unlearning operations within a short time; in the Image Captioning task, its efficiency on the COCO composite dataset is approximately $1445\times$ higher than Finetune. On average across multiple models and datasets, EMMU achieves a $68\times$ improvement. These results demonstrate that whether under the basic or enhanced unlearning setting, EMMU completes unlearning in extremely short time, showcasing its efficiency and practical value.

TABLE X
PERFORMANCE OF THE MIA-S ATTACK MODEL ON DIFFERENT DATASETS

Dataset	Acc	F1	AUC	Advantage
FIU	99.05	99.05	99.84	98.10
FOI	99.60	99.60	99.98	99.20
FCI	93.80	93.71	98.81	87.60

TABLE XI
MIA-S EVALUATION RESULTS OF DIFFERENT UNLEARNING METHODS. † DENOTES OUR EMMU METHOD

Methods	FIU		FOI		FCI	
	MPR ↓	BRS ↑	MPR ↓	BRS ↑	MPR ↓	BRS ↑
Original	49.95	0.09	50.00	0.13	48.60	0.25
Retrain	0.90	1.00	0.40	1.00	4.80	1.00
Finetune	45.40	0.19	34.20	0.48	41.80	0.46
MultiDelete [10]	49.80	0.09	50.00	0.13	48.60	0.26
KUL [8]	6.20	0.87	44.00	0.45	30.20	0.55
LLMU [9]	24.05	0.56	3.60	0.89	<u>28.40</u>	<u>0.58</u>
SalUn [33]	36.20	0.34	32.40	0.52	34.00	0.48
EMMU †	<u>12.40</u>	<u>0.76</u>	<u>27.80</u>	<u>0.60</u>	26.80	0.60

4) *MIA Results Analysis*: For the MIA-S setting, we evaluate the separability between Original model and Retrain model behaviors on FIU, FOI, and FCI datasets, with results reported in Table X. The attack model achieves over 99% Accuracy and F1 Score on FIU and FOI, with AUC approaching 100%. Even on the more challenging FCI dataset, Accuracy remains at 93.80%, indicating that the model can reliably distinguish Original model behavior from Retrain model behavior.

Table XI reports the MIA-S results of different unlearning methods. The Original model consistently exhibits significant deviation from Retrain behavior across all datasets, whereas Retrain serves as the reference for the ideal unlearning state. MultiDelete, Finetune, and SalUn remain close to the Original model on all datasets, indicating limited alignment with Retrain behavior. Although KUL achieves relatively low MPR on FIU, Table I reveals a substantial degradation in model utility during unlearning. LLMU shows relatively strong alignment with the Retrain model on FOI, but its overall utility and forget quality remain inferior to EMMU (Basic).

EMMU consistently improves alignment with Retrain behavior across all datasets while maintaining a balance between privacy and utility. On FIU and FOI, it achieves competitive performance compared with existing methods. On FCI, EMMU (Enhanced) obtains the lowest MPR and strong BRS, indicating effective suppression of behavioral deviation while preserving Retrain consistent outputs.

We further evaluate EMMU (Enhanced) under the standard MIA setting. The attack model achieves 92.00% Accuracy, 92.06% F1 Score, 97.46% AUC, and 84.00% Advantage, confirming that the attack model is effective under the standard MIA setting. As shown in Table XII, Retrain still achieves 90.40% Accuracy and 90.32% F1 Score, suggesting that the model learns distributional patterns from D_r that generalize to D_u , leaving residual membership-related signals detectable by the adversary. In contrast, EMMU (Enhanced) further reduces the attack performance to 80.80% Accuracy and 78.58% F1

TABLE XII
MIA RESULTS OF DIFFERENT UNLEARNING METHODS. † DENOTES OUR EMMU METHOD

Methods	Acc ↓	F1 ↓	AUC ↓	Advantage ↓
Retrain	90.40	90.32	96.64	80.80
EMMU (Enhanced) †	80.80	78.58	92.20	61.60

TABLE XIII
UNLEARNING PERFORMANCE UNDER DIFFERENT CRUCIAL PARAMETERS LOCALIZATION STRATEGIES

Methods	FIU			FOI		
	MU ↑	FQ ↓	UFB ↑	MU ↑	FQ ↓	UFB ↑
EMMU (Basic)	188.33	2.64	0.17	209.42	2.29	0.27
EMMU (Basic) w/o shap	186.73	2.86	0.16	208.62	2.33	0.27
EMMU (Basic) w/o input	187.23	2.72	0.16	208.94	2.45	0.26

TABLE XIV
IMPACT OF THE ALIGNMENT LOSS $\mathcal{L}_3$ ON UNLEARNING PERFORMANCE

Methods	FIU			FOI		
	MU ↑	FQ ↓	UFB ↑	MU ↑	FQ ↓	UFB ↑
EMMU (Basic)	188.33	2.64	0.17	209.42	2.29	0.27
EMMU (Basic) w/o $\mathcal{L}_3$	186.49	1.39	0.27	210.11	8.12	0.10

Score, while the AUC decreases to 92.20% and the Advantage drops to 61.60%, demonstrating that it can further weaken the separability between member and non-member samples and reduce the attack model's ability to infer membership status.

In addition, we compute the MPR and BRS metrics on the unlearning set D_u . The MPR of the Retrain model is 87.20%, while EMMU (Enhanced) reduces it to 70.80%, indicating fewer samples are classified as Original-like by the attack model. Meanwhile, the BRS changes from 1.00 for Retrain to 0.93 for EMMU (Enhanced), while still remaining at a relatively high level, suggesting that EMMU (Enhanced) maintains strong consistency with the Retrain model in the representation space.

C. Ablation Studies

1) *Impact of the Strategy Components*: To validate the necessity of both input sensitivity and SHAP-based attribution in our parameter localization strategy, we conduct ablation experiments by removing each component individually. As shown in Table XIII, removing either criterion degrades model utility and forget quality on both FIU and FOI. Since the two perspectives capture parameter importance from complementary angles, taking their intersection yields a more precise identification of neurons strongly associated with sensitive information, thereby improving unlearning effectiveness while limiting damage to model performance.

2) *Impact of the Alignment Loss $\mathcal{L}_3$* : Table XIV reports the performance with and without $\mathcal{L}_3$, illustrating its role in mitigating excessive unlearning. We observe that on FIU, removing $\mathcal{L}_3$ causes the model's outputs on the unlearning set to degenerate into repetitive "ap"; on FOI, outputs collapse into repetitive "occasional", with the FQ metric rising sharply from 2.29 to 8.12, indicating severe excessive unlearning. By aligning

TABLE XV

IMPACT OF THE NUMBER OF NEURONS R ON UNLEARNING PERFORMANCE

R	FIU			FOI		
	MU $\uparrow$	FQ $\downarrow$	UFB $\uparrow$	MU $\uparrow$	FQ $\downarrow$	UFB $\uparrow$
500	187.61	2.73	0.16	209.22	2.42	0.26
1500	188.39	2.72	0.16	209.12	2.29	0.27
2500	188.33	2.64	0.17	209.42	2.29	0.27
3500	187.94	2.85	0.16	208.93	2.38	0.26
4500	188.05	2.70	0.16	208.61	2.29	0.27

TABLE XVI

IMPACT OF DIFFERENT LOSS BALANCING FACTOR COMBINATIONS (γ_1 , γ_2 , AND γ_3) ON UNLEARNING PERFORMANCE

$(\gamma_1, \gamma_2, \gamma_3)$	FIU			FOI		
	MU $\uparrow$	FQ $\downarrow$	UFB $\uparrow$	MU $\uparrow$	FQ $\downarrow$	UFB $\uparrow$
(0.1, 1.0, 1.0)	188.33	2.64	0.17	209.42	2.29	0.27
(0.5, 1.0, 1.0)	147.48	1.10	0.25	207.76	3.37	0.20
(1.0, 1.0, 1.0)	145.64	0.34	0.45	207.24	3.95	0.18
(0.1, 0.1, 1.0)	181.13	3.04	0.14	208.13	2.90	0.23
(0.1, 0.5, 1.0)	184.10	2.68	0.16	208.34	3.34	0.20
(0.1, 1.0, 0.1)	192.87	5.41	0.09	208.76	4.74	0.15
(0.1, 1.0, 0.5)	186.39	2.71	0.16	208.05	3.48	0.20

unlearning set images with randomly sampled text from the remaining set, $\mathcal{L}_3$ imposes a semantic regularization constraint during parameter modification, effectively suppressing the generation of meaningless outputs.

3) *Impact of the Number of Neurons R* : To analyze how R affects unlearning performance, we conduct experiments under varying values of R . As shown in Table XV, the overall unlearning performance remains relatively stable across different R settings. Once R reaches a sufficient scale, the $Top-K$ selection strategy already covers most neurons strongly associated with sensitive information, so further increasing R only introduces weakly relevant parameters with marginal effect on unlearning quality and model utility. Moreover, during parameter updates in EMMU (Basic), the crucial parameter pool is dynamically updated based on the current batch, meaning that even with a large R , the number of parameters actually updated remains constrained by the input samples, preventing excessive perturbation to overall model performance. Overall, a moderate selection scale ($R = 2500$) achieves a favorable balance between unlearning effectiveness and model utility.

4) *Impact of the Loss Balancing Factors γ_1 , γ_2 , and γ_3* : Table XVI shows the effect of different balancing factor combinations on unlearning performance. When γ_1 is set large ((1.0,1.0,1.0)), the forget quality metric is low, but model utility drops significantly, with outputs degenerating into repetitive meaningless characters, indicating that over-emphasizing the unlearning objective substantially harms overall model performance. When γ_3 is too small, the model utility improves (MU=192.87), but FQ increases sharply and the overall utility-forget balance degrades (UFB=0.09), suggesting that insufficient output regularization leads to forget quality deterioration. Overall, (0.1,1.0,1.0) yields stable results on both FIU and FOI

TABLE XVII

IMPACT OF DIFFERENT CRUCIAL PARAMETER SELECTION RATIOS (α , β) IN THE ENHANCED STRATEGY ON UNLEARNING PERFORMANCE

$(\alpha\%, \beta\%)$	MU $\uparrow$	FQ $\downarrow$	UFB $\uparrow$
(90%, 10%)	207.95	3.49	0.86
(80%, 20%)	205.68	3.20	0.86
(70%, 30%)	206.36	3.68	0.85
(60%, 40%)	206.82	4.24	0.83
(50%, 50%)	209.56	3.11	0.88
(40%, 60%)	206.82	3.41	0.86

dataset, maintaining a favorable balance between model utility and unlearning effectiveness.

5) *Impact of the Weights $\alpha\%$ and $\beta\%$ in the Enhanced Strategy*: Table XVII shows the effect of different selection ratios of the two types of candidate crucial parameters in EMMU (Enhanced). Specifically, $\alpha\%$ denotes the proportion of crucial parameters selected based on input sensitivity and SHAP scores, and $\beta\%$ denotes the proportion selected based on parameter difference scores. As shown in the results, $(\alpha\%, \beta\%) = (50\%, 50\%)$ achieves the best performance across MU, FQ, and UFB. This suggests that when the contributions of the two parameter sources are kept roughly balanced, the model achieves a better trade-off between unlearning effectiveness and model utility, yielding more stable overall performance.

VII. CONCLUSION

In this paper, we present EMMU, an efficient multimodal model unlearning framework. EMMU incorporates a basic strategy for forgetting regular sensitive samples and an enhanced strategy for handling hard samples with strong memorization. By locating and updating only a small subset of crucial parameters, EMMU achieves effective removal of sensitive information while preserving overall model fidelity. Experimental results demonstrate that EMMU performs excellently across various vision-language tasks and effectively removes sensitive information in models of different sizes, while maintaining model performance fidelity. Our work offers a novel and effective solution for privacy protection in multimodal models.

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